A Rigorous Analysis of the Taxonomic Graph Analysis Personality Hierarchy and Its Application in LLM Agent Simulation

Executive Summary

For the architectural foundation of multi-agent social simulations utilizing Large Language Models (LLMs), a hybrid personality structure is currently recommended. While the 2025 Taxonomic Graph Analysis (TGA) model presents a mathematically superior, data-driven framework by introducing a distinct "Disinhibition" meta-trait, its 28 specific first-level facets and their corresponding covariance matrices remain unpublished in the available archival data. Therefore, the recommended protocol is to utilize the extensively validated 30-facet correlation matrix of the IPIP-NEO-120 to execute multivariate normal sampling for population generation, while mapping these generated profiles onto the TGA's three meta-traits (Stability, Plasticity, Disinhibition) to guide agent behavior. The inclusion of Disinhibition provides critical theoretical bandwidth for simulating impulsive, externalizing, and socially antagonistic behaviors that standard Big Five models fail to isolate. This recommendation would pivot to a strict, end-to-end TGA implementation immediately upon the open-source release of the TGA facet-level covariance matrices.

1. The Taxonomic Graph Analysis (TGA) Model of Personality

The accurate simulation of human behavior within synthetic, multi-agent environments relies heavily on the structural fidelity and psychometric validity of the underlying personality framework. Historically, personality psychology has been dominated by top-down, factor-analytic models derived from the psycholexical hypothesis. However, the 2025 research conducted by Samo, Garrido, Abad, Golino, McAbee, and Christensen introduces a paradigm shift through the application of Taxonomic Graph Analysis (TGA) to the International Personality Item Pool (IPIP-NEO)[1].

Structural Reorganization: Traits and Meta-Traits

The TGA model completely restructures the traditional personality hierarchy into a three-tiered system comprising 3 meta-traits at the third level, 6 traits at the second level, and 28 facets at the first level[2]. By allowing items to cluster based on their natural statistical relationships rather than forcing them into pre-defined theoretical silos, the model reveals substantial deviations from the classical Five-Factor Model (FFM)[3].

The Six Second-Level Traits

The TGA methodology recovered three traits that directly align with the traditional Big Five, but

it fundamentally reorganized the remaining domain space (traditionally occupied by Extraversion and Agreeableness) into three newly emergent trait-level clusters[1]:

1. **Neuroticism**: Retained from the Big Five, representing emotional sensitivity, threat detection, and reactivity to stress.
2. **Conscientiousness**: Retained from the Big Five, representing self-discipline, organization, and goal-directed behavior.
3. **Openness**: Retained from the Big Five, representing cognitive exploration, curiosity, and tolerance for novelty and aesthetics.
4. **Sociability**: A newly formulated trait representing the socially engaging, gregarious, and outgoing elements traditionally housed under Extraversion[4].
5. **Integrity**: A newly formulated trait capturing honesty and moral straightforwardness. This trait conceptually bridges elements of traditional Agreeableness and the Honesty-Humility factor isolated in the HEXACO model[4].
6. **Impulsivity**: A newly formulated trait isolated from traditional Neuroticism and low Conscientiousness, representing the tendency to act on immediate urges without forethought[4].

The Three Third-Level Meta-Traits

Historically, the Big Five domains have been summarized by two higher-order meta-traits, originally proposed by Digman and refined by DeYoung: Stability and Plasticity[6]. The TGA model expands this cybernetic framework into a tripartite structure:

1. **Stability**: Encompasses the capacity to maintain functioning, regulate emotional and behavioral disruption, and avoid threats. It is associated with long-term goal pursuit and the maintenance of homeostasis[7]. Traits like Neuroticism (inversely) and Conscientiousness load heavily here.
2. **Plasticity**: Represents the tendency to approach new opportunities, engage actively with the external environment, and exhibit cognitive and behavioral flexibility[7]. Traits like Openness and Sociability are the primary drivers of this meta-trait.
3. **Disinhibition**: A fundamentally novel meta-trait identified by the TGA framework. Disinhibition captures variance related to externalizing behaviors, self-regulatory deficits, and interpersonal antagonism[3]. In clinical psychological science, disinhibition is highly relevant as it predicts externalizing disorders (e.g., antisocial behavior, substance abuse) more strongly than impulsivity alone[10]. In the context of LLM social simulations, this meta-trait is indispensable for generating agents capable of realistic norm-violation, aggression, and conflict, providing a statistical mechanism for simulating the "Dark Tetrad" behaviors that standard Big Five models struggle to accommodate.

The First-Level Facets (Table 1)

An exhaustive review of the archival data and supplementary materials confirms that the specific names, definitions, and parent-trait mappings of the 28 first-level facets recovered by the TGA model have not been explicitly published or detailed in the available texts[12]. Therefore, attempting to list them would require unsupported speculation. The table below represents the verified hierarchical mapping of the TGA model, explicitly marking the 28 facets as an open,

unverified problem pending further data releases.

Third Level (Meta-Traits)	Second Level (Traits)	First Level (Facets)	Definition / Construct Relevance
Stability	Neuroticism	*Unverified* (Subset of 28)	Emotional reactivity and threat detection.
Stability	Conscientiousness	*Unverified* (Subset of 28)	Self-discipline, organization, and order.
Plasticity	Openness	*Unverified* (Subset of 28)	Cognitive exploration and aesthetic tolerance.
Plasticity	Sociability	*Unverified* (Subset of 28)	Social engagement, outgoingness (formerly Extraversion).
Stability / Plasticity	Integrity	*Unverified* (Subset of 28)	Honesty, moral compliance (formerly Agreeableness/Honesty-Humility).
Disinhibition	Impulsivity	*Unverified* (Subset of 28)	Lack of forethought, externalizing behavior, self-regulation deficit.

(Note: While the exact loading of Integrity onto a single meta-trait is complex, it heavily influences Stability via cooperation, while Impulsivity loads onto the novel Disinhibition meta-trait.)

Methodological Innovations: Taxonomic Graph Analysis Explained

Taxonomic Graph Analysis (TGA) is a comprehensive network psychometrics framework designed to identify hierarchical psychological structures from the bottom up, avoiding the theoretical assumptions baked into classical models[3]. It differs fundamentally from traditional exploratory factor analysis (EFA) or principal component analysis (PCA), which operate top-down by extracting factors that maximize explained variance until no meaningful variance remains[3].

Traditional factor analysis struggles profoundly with large item pools due to two major psychometric artifacts: local dependence and wording effects[2]. Local dependence occurs when two items are highly correlated merely because they share synonymous phrasing, not because they measure the same construct. Wording effects occur when respondents exhibit bias toward positively or negatively phrased items, known as acquiescence bias. For example, in the traditional IPIP-NEO, the items "Am afraid to draw attention to myself" (Neuroticism), "Don't like to draw attention to myself" (Extraversion), and "Dislike being the center of attention" (Agreeableness) artificially inflate cross-trait correlations due to shared syntax rather than shared underlying psychological reality[3].

TGA mitigates these artifacts through a rigorous, multi-step algorithm:

1. **Redundancy Removal**: The algorithm first identifies and removes items that violate local independence, ensuring that semantic similarity does not masquerade as psychological correlation.
2. **Wording Effect Mitigation**: TGA fits a random intercept factor model to absorb the variance caused by response styles (e.g., the tendency to agree with negatively worded statements). The residual correlation matrix—cleansed of this artifact—is then used for the actual structural analysis[3].
3. **Exploratory Graph Analysis (EGA)**: Rather than extracting factors, EGA constructs a network where items are represented as nodes, and the partial correlations between them serve as edges.
4. **Louvain Community Detection**: A network algorithm groups highly connected nodes into specific communities, which form the facets and traits[3].
5. **Bootstrapping for Stability**: The process is repeated across 500 simulated samples to calculate "item stability." Items that do not land in their designated cluster at least 75% of the time are considered psychometrically unstable and are removed[3].

Sample Parameters and Robustness

The TGA analysis utilized massive archival personality data from the Johnson IPIP-NEO repository, one of the largest public-domain personality samples available, featuring $N = 320,128$ individuals from the United States[3]. By utilizing bootstrapped EGA with 500 replicates and enforcing the strict 0.75 item stability threshold, the authors ensured that the resulting 28-facet, 6-trait structure represents a highly robust psychometric consensus rather than a sample-specific artifact[3].

2. Comparative Structural Analysis of Personality Models

To evaluate the TGA model against established frameworks for the purpose of multi-agent simulation, it is necessary to contrast its theoretical underpinnings and mathematical outputs with the dominant models in the field. Table 2 summarizes these comparative structures.

Table 2: Comparison of Established Personality Structures

Taxonomy	Levels	Traits	Facets/ Aspects	Derivation Method	Main Instrument	Strengths for Simulation	Weaknesses for Simulation
Big Five / NEO-PI-R [cite: 3, 14]	3	5	30	Lexical & Theory-driven	NEO-PI-R / IPIP-NEO-120	Vast archives of predictive validity data; standard baseline for LLM tests.	Top-down silos obscure item-level nuance; wording effects inflate cross-correlations.
Big Five Aspects [cite: 14]	3	5	10 (Aspects)	Data-driven (Factor Analysis)	BFAS-40	Bridges the gap between broad domains and extremely narrow facets.	10 aspects lack the extreme granularity needed for highly specific, individualized agent behavio

							ral generation.
HEXACO [cite: 3, 9]	3	6	24	Lexical (Cross-cultural)	HEXACO-PI-R	Isolates Honesty-Humility; excellent for simulating dark tetrad, manipulative, and adversarial behaviors.	Debated necessity of the sixth factor; less archival correlation data compared to the standard Big Five.
The Big Two [cite: 6]	1	2 (Meta)	N/A	Factor Analysis	Various	Highly parsimonious; maps perfectly to neurological approach/avoidance cybernetic systems.	Far too abstract; impossible to generate distinctly nuanced agent personas from only two variables.
Facet	2	5	70	Bottom	Facet	The	70

MAP [15]				-up (Factor Analysis)	MAP	most comprehensive assessment to date (Irwing et al., 2024); extreme behavioral granularity.	dimensions introduce high computational complexity, massive multicollinearity, and overfitting in simulations.
TGA Hierarchy [cite: 1, 3]	3	6	28	Bottom-up (Network Psychometrics)	TGA-derived IPIP-NEO	Mathematically mitigates wording effects; introduces "Disinhibition" for superior externalizing modeling.	Facet-level identities and precise covariance matrices are currently unpublished, making immediate pure implementation impossible.

The comparative analysis reveals that while the Big Five remains the dominant framework due to data availability, models derived from the bottom up—such as the Facet MAP and TGA—offer

superior granularity. However, the Facet MAP's 70 dimensions present severe multicollinearity risks for statistical sampling. The TGA model strikes an optimal balance with 28 facets, provided the covariance matrix becomes accessible.

3. Critique, Limitations, and Evidence Quality

Independent Consensus and Methodological Critique

The reception of bottom-up, network-derived hierarchies like TGA among independent psychometricians reveals a bifurcated consensus. Proponents argue that traditional top-down methods enforce "mono-operation bias" and obscure meaningful behavioral differences by aggregating them into massive, artificially enforced theoretical silos[3]. They suggest TGA's identification of traits like Sociability and Integrity provides a more precise map of human nature and could transform psychopathology classifications, moving away from rigid diagnostic categories toward dimensional trait models[18].

Conversely, critics of expanding the Big Five framework caution against naive interpretations of new traits. Researchers like Soto have historically demonstrated problems in finding rigorous, cross-sample evidence for six-factor models, arguing that they often capture statistical artifacts or sample-specific anomalies rather than universal human dimensions[19].

Known Limitations of the TGA Model

1. **Sample and Cultural Dependence**: The TGA model was derived from a predominantly US-based, English-speaking sample of 320,128 individuals[14]. Personality structure is known to shift across languages and cultural contexts. For example, LLMs and humans tested in different languages exhibit varied baseline traits—such as significantly lower neuroticism in Chinese cohorts, and varying openness in French cohorts[20]. The universality of the TGA traits remains unproven.
2. **Item-Pool Dependence**: Bottom-up methods are entirely constrained by the semantic breadth of the items fed into them. TGA utilized the IPIP-NEO pool; applying Taxonomic Graph Analysis to the HEXACO item pool would likely yield a fundamentally different graphical structure, highlighting that the model describes the *item pool* as much as it describes human nature[3].
3. **Self-Report Bias**: While TGA algorithmically mitigates wording effects and acquiescence bias through random intercept modeling, fundamental self-report bias (social desirability, lack of self-insight) remains a structural limitation of all underlying psychometric item pools[3].
4. **Stability of Facet-Level Structure**: While TGA enforces an item stability threshold of 0.75 via bootstrapping, facet-level constructs are historically more volatile across diverse populations than the broader meta-traits. Replicating the exact 28 facets in a different demographic is statistically challenging[3].

The Predictive Superiority of Facets

A critical question for simulation design is whether to specify agent personalities at the trait level (e.g., 6 variables) or the facet level (e.g., 28-30 variables). There is robust empirical evidence that facet-level scores predict real-world outcomes significantly better than broad,

trait-level domains. Researchers such as Paunonen, Ashton, and Mõttus have demonstrated that outcome-related information is often contained in the specific, unique variances of narrow trait measures[22]. For example, facets are superior predictors of romantic outcomes, body mass index, and specific job performances[22].

For LLM agents, defining a persona purely by "High Extraversion" results in generic, stereotyped text generation. Specifying facet-level divergence—such as "High Assertiveness but Low Gregariousness"—yields a highly specific, nuanced behavioral phenotype that reacts realistically to varied simulated stimuli.

Overall Assessment of the TGA Structure

Compared to the heavily settled Big Five and HEXACO models, the TGA structure remains in its infancy. While mathematically superior in handling local independence and wording effects, the model lacks the decades of cross-cultural validation enjoyed by the FFM[8]. For immediate engineering applications, the structural logic of TGA—specifically the inclusion of the Disinhibition meta-trait—is highly valuable, but the model cannot fully replace the Big Five backbone until its 28-facet covariance matrix is published and independently replicated.

4. Correlation Structure and Multivariate Profiling

For a multi-agent social simulation to mirror human populations realistically, agents cannot be generated by sampling personality facets independently. Psychological facets are highly intercorrelated; simulating them as orthogonal variables will yield biologically and psychologically impossible synthetic agents (e.g., an agent with maximum depression and vulnerability, but zero anxiety, which violates the covariance of the Neuroticism domain).

The Necessity of Multivariate Sampling

To generate a realistic synthetic population, personality profiles must be sampled using a multivariate normal distribution anchored by a valid empirical correlation matrix. Because the specific 28-facet correlation matrix for the 2025 TGA model is currently unpublished, the rigorously validated approach is to utilize the comprehensive 30-facet correlation matrix from the public domain IPIP-NEO-120 (Kajonius & Johnson, 2019)[26].

Sampling from a multivariate normal distribution requires a vector of facet means (μ) and a covariance matrix (Σ). By executing operations such as $\mathbf{X} \sim \mathcal{N}(\mu, \Sigma)$, the simulation ensures that the synthetic population respects the natural collinearity of human traits. For instance, the empirical matrix enforces the strong positive correlation between Gregariousness and Warmth, and the strong negative correlation between Trust and Angry Hostility[26]. Independent sampling ignores these relationships, resulting in chaotic and contradictory agent behavioral logic.

Population Demographics and Realistic Archetypes

A population-realistic simulation must also adjust the mean vector (μ) to account for demographic variance. Empirical data demonstrates distinct sex and age differences at the facet level that simulations must respect:

- **Sex Differences**: Females generally score higher on Agreeableness and specific facets of

Neuroticism (Anxiety, Vulnerability) compared to males, while males often score higher on assertiveness[19].

- **Age Differences**: Openness tends to peak in early adulthood and stabilize or slightly decline with advanced age, while Conscientiousness generally increases linearly across the lifespan[19].

When continuous multivariate data is subjected to density clustering, it rarely yields highly distinct, isolated "types," as personality exists on a continuum. However, density analyses consistently identify stable profiles or archetypes, often categorized cybernetically as:

1. **The Resilient**: Characterized by high Stability and high Plasticity.
2. **The Overcontrolled**: Characterized by high Stability but low Plasticity (highly cautious, rule-bound).
3. **The Undercontrolled**: Characterized by low Stability and high Disinhibition (prone to externalizing, impulsive behavior)[19]. Other less formal clustering methods label these density peaks as Role Model, Self-centered, Reserved, and Average[19]. An effective simulation should naturally produce these archetypes as emergent properties of the covariance matrix.

5. Application to LLM Agents

Default Personalities and the Alignment Problem

Recent empirical studies subjecting Large Language Models to psychometric inventories—such as the TRAIT benchmark, PsychoBench, and PersonaLLM—reveal that unprompted LLMs exhibit highly consistent, distinct personalities largely shaped by their training data and alignment tuning (e.g., RLHF or RLAIF)[28]. Current state-of-the-art models systematically default to a specific psychometric profile: extremely high Agreeableness, high Conscientiousness, and exceptionally low Neuroticism[29].

This alignment results in "safe," calm, emotionally stable, and highly compliant text generation. However, if all agents in a social simulation share this default profile, the simulation will completely fail to generate realistic conflict, deviance, exploitation, or complex social dynamics[30]. Therefore, active "steering" of the LLM's personality is mandatory.

Facet-Level Agent Specifications and Prompting

Research indicates that prompting LLMs at the facet level yields significantly better behavioral adherence than domain-level prompting[31]. The benchmark *PersonalityEdit* demonstrates that utilizing specific facet descriptors (e.g., embedding words related to specific sub-traits like vulnerability or assertiveness) allows models to bypass their default alignment and express targeted traits[31].

To build a compact, effective agent description from a sampled 30-dimensional profile, the following protocol is recommended:

1. **Filter by Extremes**: Do not feed the LLM all 30 facets in the system prompt. LLMs suffer from prompt dilution and context degradation. Select only the facets where the statistically sampled score falls in the top or bottom decile (e.g., $Z > 1.5$ or $Z < -1.5$).

2. **Discard Mid-Range Traits**: Traits near the population mean ($Z \approx 0$) should be omitted from the prompt, as the LLM's default baseline behaviors generally suffice for average interactions.
3. **Translate to Natural Language**: LLMs process semantic descriptors better than raw numerical scores. Convert the standard scores into "sufficient statistic" narrative descriptors (e.g., rather than "Extraversion: 8/10", use "You have a highly assertive and gregarious personality; you prefer sociable hobbies to quiet, solitary ones and seek external stimulation")[28].

Validation Protocol for LLM Agents

To ensure the LLM successfully internalizes the prompted profile, the agent must be periodically evaluated using established psychometric protocols during the simulation.

1. **Instrument Selection**: Utilize the IPIP-NEO-120 or the BFI-2, enhanced with situational knowledge graphs to test behavioral intent rather than simple self-awareness (e.g., presenting a scenario rather than asking "Are you outgoing?")[28].
2. **Forced-Choice Architecture**: Human personality tests adapted to LLMs often suffer from severe social desirability bias, where the LLM refuses to express negative traits. To counter this, validation should utilize forced-choice testing paradigms where the LLM is presented with two statements of equal social desirability and must choose one[20]. This methodology drastically increases the internal validity and reliability of the assessment.
3. **Tolerance Metrics**: A successfully expressed profile should yield tested facet scores within ±1 Likert point (or falling within the 95% Confidence Interval) of the injected initialization profile across varying temperature settings[20].

6. Implementation Recipe: Generating Realistic Agent Profiles

The following step-by-step recipe and Python implementation demonstrate how to generate realistic personality profiles using numpy by sampling from a predefined correlation matrix[27].

Step 1: Obtain the Covariance Matrix Locate the published 30x30 facet correlation matrix from the Kajonius & Johnson (2019) IPIP-NEO-120 study[26]. Convert this correlation matrix into a covariance matrix by multiplying by the standard deviations of each facet.

Step 2: Define Population Means Define the 1x30 vector of population means, adjusting slightly based on the desired demographic distribution of your synthetic agents (e.g., skewing certain facets based on simulated age or sex).

Step 3: Multivariate Sampling Execute the Python code below to generate the population, ensuring collinearity is preserved.

```python
import numpy as np
```

```
def generate_population_profiles(num_agents, facet_means, covariance_matrix):
    """
    Generates a population of realistic personality profiles using multivariate normal sampling.

    Parameters:
    num_agents (int): The number of synthetic agents to generate.
    facet_means (array): A 1D array of length N containing the population means for each facet.
    covariance_matrix (array): An N x N positive semi-definite covariance matrix.

    Returns:
    np.ndarray: An array of shape (num_agents, N) containing agent profiles.
    """
    # Sample from the multivariate normal distribution to preserve collinearity
    # (e.g., ensuring high warmth correlates with high gregariousness)
    profiles = np.random.multivariate_normal(facet_means, covariance_matrix, num_agents)

    # Clip the values to realistic Likert-scale bounds (e.g., 1.0 to 5.0)
    profiles = np.clip(profiles, 1.0, 5.0)

    return profiles

# Example usage (simplified to 5 facets for brevity; expand to 30 for IPIP-NEO)
# Means for 5 facets based on a theoretical 1-5 scale
means = np.array([3.2, 2.8, 3.5, 3.1, 2.9])

# A sample covariance matrix demonstrating positive/negative correlations
cov_matrix = np.array([
    [1.0, 0.6, 0.4, 0.1, 0.0],
    [0.6, 1.0, 0.2, 0.0, 0.1],
    [0.4, 0.2, 1.0, 0.3, 0.1],
    [0.1, 0.0, 0.3, 1.0, 0.5],
    [0.0, 0.1, 0.1, 0.5, 1.0]
])

# Generate 1000 realistic agents
agent_population = generate_population_profiles(1000, means, cov_matrix)
```

7. Open Problems and Unverifiable Claims

Despite the robust framework of the TGA analysis, several critical gaps exist in the currently available literature that restrict an immediate, pure 1-to-1 deployment of the 2025 model:

1. **The 28 Facets**: As explicitly requested by the parameters of this research, the precise names, definitions, and parent-trait mappings of the 28 first-level facets derived in the Samo et al. (2025) TGA model are completely absent from the provided source material

and public repositories. It is explicitly noted that this information is currently unverifiable[12].

2. **The TGA Correlation/Covariance Matrix**: To map agent personalities mathematically via TGA, a 28x28 covariance matrix is required. This matrix has not been published in the available archival text or linked OSF repositories. Consequently, the established 30-facet IPIP-NEO correlation matrices must serve as a temporary proxy.
3. **Cross-Cultural Facet Stability**: While the structural stability of the traits is established for the US-based English dataset ($N = 320,128$), the exact stability of the specific TGA facets across non-Western cultures and languages remains an open problem[14].

8. Reference List and Claim Verification

- [1]
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  - *Claim Support*: **Strongly Supported**. Outlines the TGA methodology, the mitigation of wording effects, the 6 traits, and the 3 meta-traits.
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  : Irwing, P., Hughes, D. J., Tokarev, A., & Booth, T. (2024). "Towards a taxonomy of personality facets." *European Journal of Personality*.
  - *Claim Support*: **Strongly Supported**. Outlines the 70-facet bottom-up Facet MAP structure.
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  : Kajonius, P., & Johnson, J. A. (2019). "Assessing the structure of the Five Factor Model of Personality (IPIP-NEO-120) in the public domain." *Europe's Journal of Psychology*.
  - *Claim Support*: **Strongly Supported**. Provides the 30-facet correlation matrices required for multivariate sampling.
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  - *Claim Support*: **Strongly Supported**. Demonstrates that facet-level variance predicts real-world behavioral outcomes superior to trait-level variance.
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  - *Claim Support*: **Strongly Supported**. Documents LLM default traits (High Agreeableness/Conscientiousness, Low Neuroticism), prompt steering mechanics, and the efficacy of forced-choice testing over Likert scales to defeat social desirability bias in LLMs.
- [3]
  : Various clinical psychological sources regarding approach/avoidance and externalizing.
  - *Claim Support*: **Mixed**. Defines "Disinhibition" as distinct from standard impulsivity, specifically regarding its linkage to externalizing disorders, though the exact boundaries

between Disinhibition and low Conscientiousness/Agreeableness remain debated in the broader literature.

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